An atomistic model for surface corrosion of RuO_2-based electrocatalysts for the acidic oxygen evolution reaction

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Introduction

The shift to sustainable hydrogen production via water electrolysis relies on overcoming stability issues in high-performance OER catalysts. Ruthenium dioxide (RuO_2) shows excellent activity but suffers from Ru dissolution, leading to stability loss as shown in **Figure 1a**. To address this, we use Density Functional Theory (DFT) to study the effects of dopants on the stability of RuO_2 surfaces, focusing on the (110) surface orientation. We identify those certain dopants, like Ir and Ti, enhance corrosion resistance by modifying reaction pathways and stabilizing surface sites. Furthermore, we demonstrate a strong correlation between the Gibbs free energy of RuO_4 formation (ΔG_{tot}) and the highest intermediate energy (ΔG_{max}), establishing ΔG_{tot} as a reliable descriptor for screening stability enhancements.

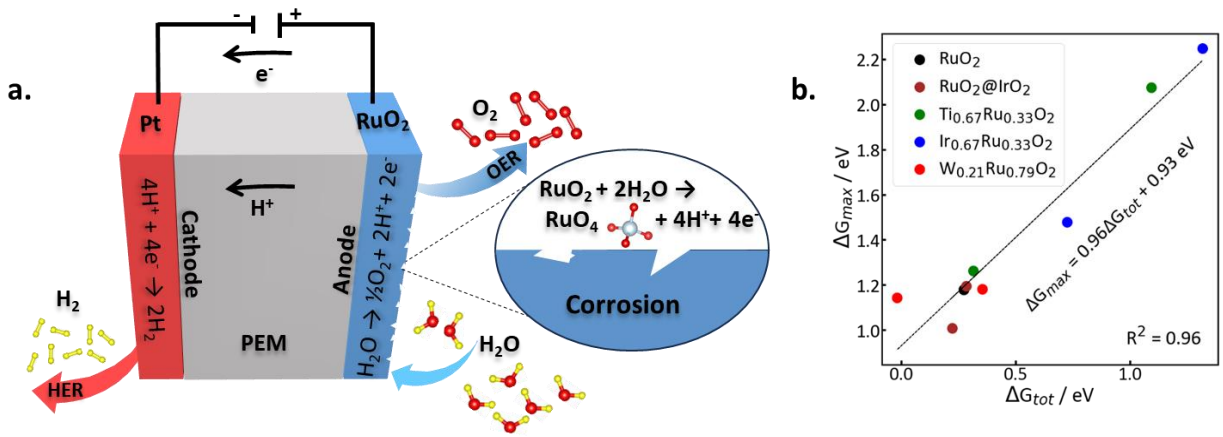

Figure 1. a) Polymer electrolyte membrane (PEM) electrolyzer. b) Correlation between ΔG_{tot} and ΔG_{max} illustrates material stability interconnections across surfaces. Free energies were calculated at pH = 0 and U_{SHE}=1.23 V.

Results and Discussion

Our study investigates RuO_2 corrosion by examining overall site stability against RuO_4 formation and the specific pathways leading to Ru dissolution. For RuO_2(110), Ru_{br}-O sites exhibit the highest stability (U_{corr} = 1.488 V), while Ru_{cus} sites are more susceptible to corrosion (U_{corr} ≈ 1.21 V), as illustrated in **Figure 2**. Doping with Ir, Pt, and Ti enhances stability, with Ir achieving the highest stability (U_{corr} up to 1.492 V) when optimally positioned. The substrate effects indicate that minimal strain from IrO_2 improves stability, whereas higher strain from TiO_2 diminishes it, guiding choices for doping and substrate materials to enhance RuO_2 stability in OER applications.

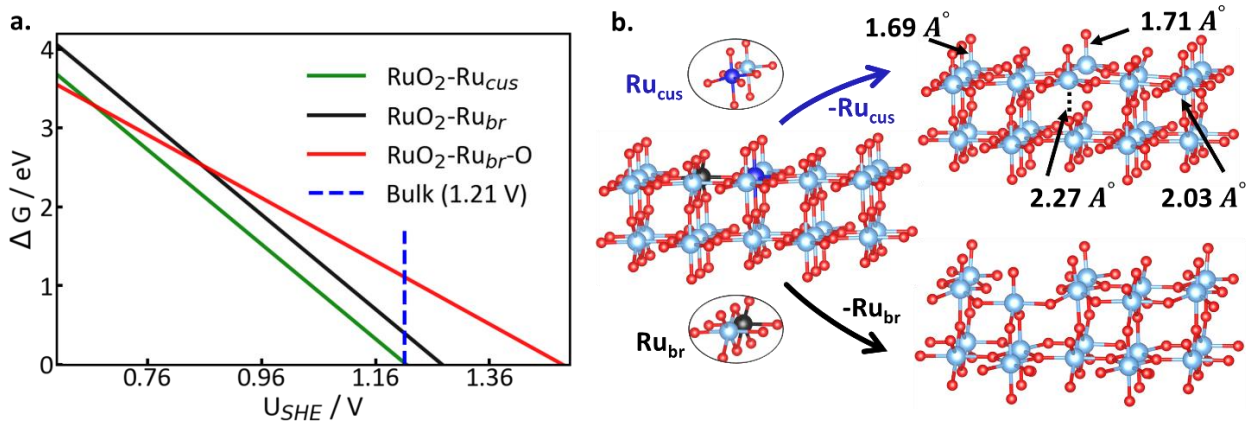

Figure 2. Thermodynamic corrosion potentials and structures of a, b): different Ru sites at the (110) surface. Colors in b): sky blue: Ru_{bulk}, blue: Ru_{cus}, black: Ru_{br}, red: O.

Furthermore, the investigation into reaction pathways for RuO_4 formation reveals that dopants like Ti, Ir, and W significantly influence the corrosion mechanism. Specifically, Ti-doped RuO_2 hinders the removal of RuO_4, while W-doped RuO_2 strongly prefers terminal O, requiring additional H_2O adsorption steps. A robust correlation (R^2 = 0.96) between total free energy and maximum free energy across various configurations establishes total free energy as a valuable descriptor for evaluating the stability of modified RuO_2 surfaces, as shown in **Figure 1b**. These findings underscore the importance of careful dopant selection in optimizing corrosion resistance.

References

1. F. Hess ACS Catal. 13 (2023), 3433.
2. W. Liu et al. J. Phys. Chem. C 127 (2023), 5334–5342.